

Q- Perform a clean OS installation with Manual disk partitioning and startup optimization.

Success criteria: system passes post , all Ips are reachable via ping and the os boot without errors

To perform a clean installation of an operating system by manually creating disk partitions and optimizing the system startup for better performance and reliability.

Step 1: Prepare Bootable Media

- 1. Create a bootable USB drive using Windows Media Creation Tool or Rufus.
- 2. Insert the USB drive into the computer.

Step 2: Configure BIOS/UEFI

- 1. Restart the computer.
- 2. Enter BIOS/UEFI by pressing **F2, Delete, or F12**.
- 3. Set the USB drive as the first boot device.
- 4. Save the settings and restart.

Step 3: Install the Operating System

- 1. Boot from the USB drive.
- 2. Select the language, time, and keyboard layout.
- 3. Click **Install Now**.
- 4. Accept the license agreement.
- 5. Select **Custom: Install Windows only (Advanced)**.

Step 4: Manual Disk Partitioning

- 1. Delete existing partitions (after backing up important data).
- 2. Create new partitions manually.

Partition	Size
System (C:)	100 GB
Data (D:)	Remaining Disk Space

- ☐ Format the partitions.
- ☐ Select the **C:** drive and click **Next** to begin the installation.

Step 5: Complete Installation

1. Wait for Windows to install.
2. The computer will restart automatically.
3. Complete the initial setup by creating a user account and configuring regional settings.

Step 6: Install Drivers

1. Open **Device Manager**.
2. Install all required device drivers.
3. Confirm that no devices show warning symbols.

Step 7: Startup Optimization

1. Open **Task Manager** (**Ctrl + Shift + Esc**).
2. Select the **Startup** tab.
3. Disable unnecessary startup applications.
4. Keep only essential startup programs enabled.
5. Restart the computer

Observation

- The operating system was installed successfully.
- Manual disk partitioning was completed.
- Device drivers were installed correctly.
- Startup applications were optimized for faster boot performance.
- The system operated smoothly after installation.

Success Criteria

The practical is considered successful when:

- The system successfully passes the **Power-On Self-Test (POST)** without hardware errors.
- All configured IP addresses are reachable using the **Ping** command.
- The operating system boots successfully without startup errors.
- Manual disk partitions are created and recognized correctly.
- Essential device drivers are installed and functioning properly.
- Startup optimization improves the system's boot performance.

A clean operating system installation was successfully completed using manual disk partitioning. The system passed the POST process, all configured IP addresses were reachable through the **Ping** command, and the operating system booted without any errors. Therefore, the system met all the specified success criteria.